SECTION 3 · SUBSECTION 3.8 (CONT.)
STORE, QUEUE AND WORKER

3.8.4 INTRODUCTION.

Job state lives in one of two backends behind a single *JobStore* interface. When *REDIS_URL* is set and reachable, records are JSON strings in Redis, the job list is a sorted set, and an indexed set tracks active jobs. Otherwise the same interface uses a module-level dictionary and a lock. The one idea the reader must leave with is that Redis mode is not merely persistence: it adds a concurrency protocol that memory mode does not need (figure 3.8-3).

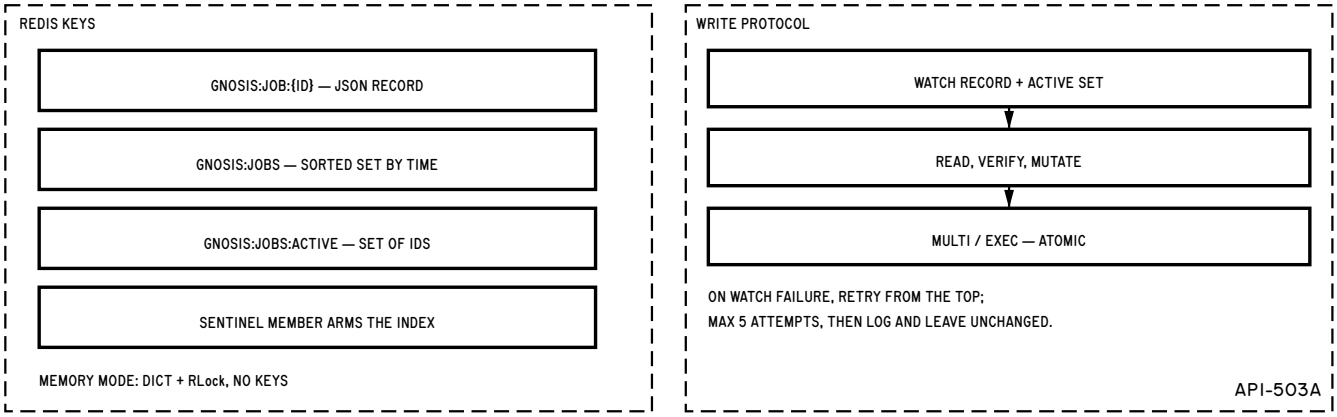


Figure 3.8-3. Redis Key Layout and Optimistic Write Loop

3.8.5 FUNCTIONAL DESCRIPTION.

create builds a queued record, writes it, adds its identifier to the list and, in Redis mode, to the active set. *start* and every phase update re-read the record inside a WATCH transaction, refuse to modify a terminal record, update fields, and remove the identifier from the active set on finish, fail or delete. If a competing writer changes the watched keys, the loop retries up to five times and then logs an error without partial writes. *active_count* reads the set cardinality, excluding the sentinel, and falls back to scanning records when the index is unarmed.

3.8.5.1 Index Reconciliation.

The active index can drift if a record expires, a process dies between two writes, or an operator clears a key. *rebuild_active_index* scans job records, rewrites the set, and arms it only after a complete pass. The migration command *job_store_migrate.py* refuses to arm unless the operator passes *--arm --confirm-all-writers-updated*; this prevents an old writer that does not maintain the set from corrupting the count.

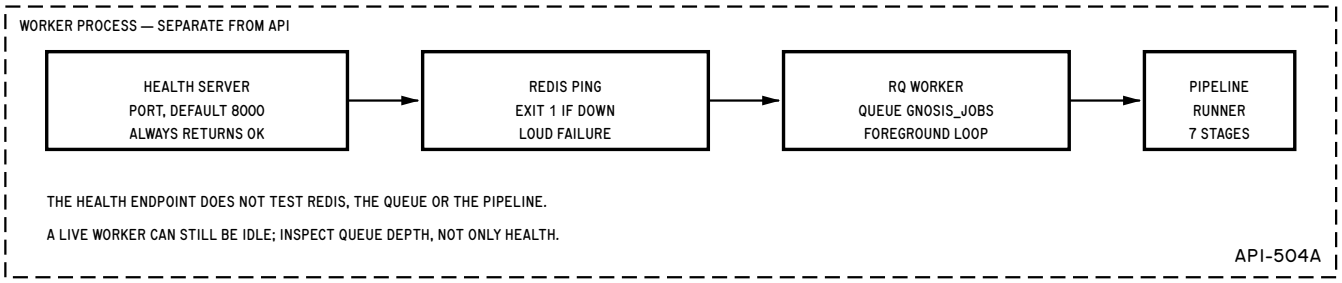


Figure 3.8-4. Worker Process and Its Health Surface

3.8.6 PERFORMANCE AND DESIGN DATA.

Table 3.8-2. Store, Queue and Worker Settings

Setting	Default	Effect
REDIS_URL	unset	Selects memory mode when absent
RQ_QUEUE_NAME	gnosis_jobs	Queue consumed by worker
JOB_TIMEOUT_SECONDS	1800	RQ execution limit
JOB_TTL_SECONDS	86400 / 0	RQ result TTL; store record TTL is 0
WATCH_MAX_ATTEMPTS	5	Optimistic write retries
ACTIVE_INDEX_SENTINEL	fixed string	Marks an armed index
Worker PORT	8000	Health listener
Local executor	2 threads	Fallback concurrency

CAUTION

Queue results expire after 86,400 seconds, but completed job records have no expiry unless *JOB_TTL_SECONDS* is also set in the API process. With the documented default the two layers diverge: the record survives while the RQ result is gone. Choose one retention policy.

```
STORE CHECK                                  OPERATOR
1 READ - /health queue_mode
2 CONFIRM - redis_connected true
  √ DEPTH - worker draining
3 CREATE - job reaches running
4 VERIFY - active count returns to zero
```